

SEE COMPULSORY SCIENCE & TECHNOLOGY

Master Solution Compendium & Evaluation Guide

Specification Grid Alignment: National Examination Board (NEB) 2080 / Set C

Total Marks: 75 | **Time Allocation:** 3 Hours

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Section 'A' — Multiple Choice Questions (10 × 1 = 10 Marks)

(a) Which of the following represents an analog signal?

Correct Choice: (iv) Short wave radio

Reasoning: Short-wave radio transmissions utilize continuously varying electromagnetic wave amplitudes or frequencies, representing pure analog signatures unlike the quantized digital states found in computers or smartwatches.

(b) Why is a snail kept in the Mollusca phylum?

Correct Choice: (i) Body is divided into head, visceral mass, muscular foot and mantle.

Reasoning: These distinctive morphological features define phylum Mollusca. Snails possess soft, unsegmented bodies enclosed by a protective mantle that secretes a calcareous shell structure.

(c) Shyam saw two invertebrates, P and Q. P is the first multicellular animal and Q is the first tissue-graded animal. Which statement matches?

Correct Choice: (ii) Body of P bears pores whereas body of Q bears coelenteron.

Reasoning: Organism P belongs to phylum Porifera (sponges) which possess cellular-level structures with porous ostia. Organism Q belongs to phylum Coelenterata (Cnidaria), which has a localized coelenteron cavity.

(d) Which part of calamus (■■■■) is used for medicinal purposes?

Correct Choice: (i) Rhizome

Reasoning: The underground modified stem or rhizome of *Acorus calamus* contains essential oils (like asarone) highly valued for mitigating respiratory issues and localized gastrointestinal distress.

(e) What would be the weight of a body of mass 20 kg at a distance of 12000 km from the centre of the Earth? ($M = 6 \times 10^{24}$ kg)

Correct Choice: (iii) 55 N

Reasoning: Yielded through Newton's Law of Gravitation combined with weight calculations:

$$g = (G \times M) / R^2 = (6.67 \times 10^{-11} \times 6 \times 10^{24}) / (12 \times 10^6)^2 \approx 2.779 \text{ m/s}^2.$$

Weight (W) = $m \times g = 20 \text{ kg} \times 2.779 \text{ m/s}^2 \approx 55.58 \text{ N}$, making 55 N the closest physical matching alternative.

(f) A ship going from sea to river has to displace more water, why?

Correct Choice: (iii) Maintain the same buoyant force

Reasoning: To stay afloat, a ship must displace a weight of water equivalent to its own total weight. Because freshwater in a river is less dense than saline seawater, a greater spatial volume must be displaced to generate an equivalent upthrust.

(g) Study the given prism dispersion diagram and identify colors A and C:

Correct Choice: (iv) A - (Red) ■■■■, C - (Violet) ■■■■

Reasoning: During light dispersion through a glass prism, red light (A) refracts the least due to its long wavelength and higher propagation velocity inside glass, remaining at the upper boundary. Violet light (C) undergoes the highest refraction.

(h) What is the value of Hubble's constant (H)?

Correct Choice: (iv) 73 km/s/Mpc

Reasoning: Modern astrophysical telemetry and contemporary standard curricula universally fix the localized value of Hubble's expansion constant at approximately 73 km/s/Mpc.

(i) If X is a metal, what is the valency of X in the reaction: $X + 2HCl \rightarrow XCl_2 + H_2$?

Correct Choice: (ii) 2

Reasoning: A singular atom of metal X combines directly with two monovalent chlorine atoms (Cl^-) to produce a stable salt, confirming an implicit chemical valency of 2.

(j) Organic matter from food wastes is a major pollutant. Why is excess organic matter a threat to aquatic life?

Correct Choice: (ii) Microorganisms consume dissolved oxygen to decompose organic matter.

Reasoning: Aerobic decomposers consume large volumes of dissolved oxygen (DO) to break down organic pollutants, causing a surge in Biochemical Oxygen Demand (BOD) that suffocates local aquatic organisms.

Section 'B' — Very Short Answer Type Questions ($9 \times 1 = 9$ Marks)

a. In an experiment, density of water is measured at different temperatures. Which one is the dependent variable?

Answer: The **density of water** is the dependent variable, while the temperature is intentionally varied as the independent factor.

b. Write Archimedes' Principle.

Answer: Archimedes' Principle states that when a body is partially or completely immersed in a fluid, it experiences an upward buoyant force (upthrust) equal to the total weight of the fluid displaced by the body.

c. Write a difference between an incident ray and an emergent ray.

Answer: An **incident ray** represents the initial ray of light striking a refracting medium interface from an external origin, whereas an **emergent ray** is the final beam exiting the medium back into its surroundings.

d. Ravi's mother reads a newspaper by keeping it more than 25 cm from her eyes. What kind of lenses should she wear?

Answer: She should wear spectacles integrated with **Convex lenses** to correct Hypermetropia (long-sightedness) by focusing nearby rays directly onto the retina.

e. What change occurs in the output voltage of a transformer if the number of turns in the secondary coil is less than the primary coil? Why?

Answer: The output voltage **decreases** because it functions as a step-down transformer. Output voltage directly correlates with the turns ratio: $V_s/V_p = N_s/N_p$. Since $N_s < N_p$, V_s will be lower than V_p .

f. Write a difference between a closed universe and an open universe.

Answer: A **closed universe** contains enough mass density to eventually reverse expansion through gravitational pull, leading to a collapse ('Big Crunch'). An **open universe** lacks this density and will expand infinitely.

g. At what temperature does the density of water become the highest?

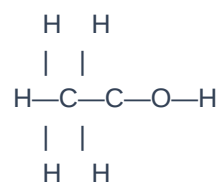
Answer: Water achieves its maximum density of 1 g/cm^3 (1000 kg/m^3) at precisely **4°C** due to its unique anomalous expansion characteristics.

h. Why is hematite ore considered the chief ore of iron?

Answer: Hematite (Fe_2O_3) is the chief ore because it boasts a high iron yield (up to 70%) and fewer complex impurities, making it highly cost-effective for industrial extraction.

i. Write the structural formula of ethanol.

Answer: The structural layout is:



Section 'C' — Short Answer Type Questions (14 × 2 = 28 Marks)

Q3. Check the correctness of the equation $v^2 = u^2 + 2as^2$ by using unit-wise analysis.

Solution:

1. Define the respective SI units for all terms:

- Velocity (v, u) = $\text{m} \cdot \text{s}^{-1} \rightarrow$ Unit of v^2 and $u^2 = (\text{m} \cdot \text{s}^{-1})^2 = \text{m}^2 \cdot \text{s}^{-2}$.
- Acceleration (a) = $\text{m} \cdot \text{s}^{-2}$.
- Displacement (s) = $\text{m} \rightarrow$ Unit of $s^2 = \text{m}^2$.

2. Substitute these units into the expression $2as^2$ (omitting the dimensionless constant 2):

- Unit of $(as^2) = (\text{m} \cdot \text{s}^{-2}) \times (\text{m}^2) = \text{m}^3 \cdot \text{s}^{-2}$.

3. Comparison: LHS Unit ($\text{m}^2 \cdot \text{s}^{-2}$) $\neq$ RHS Term Unit ($\text{m}^3 \cdot \text{s}^{-2}$).

Conclusion: The equation is dimensionally inconsistent and mathematically **incorrect**.

Q4. Write any two characteristics of monocotyledon plants.

1. Their seeds possess a **single embryonic cotyledon** (seed leaf).
2. They typically feature a **fibrous root system** and leaves displaying parallel venation patterns.

Q5. Study the concept map and mention the number of chromosomes in X and Y to distinguish haploid and diploid.

- **X (Fertile Queen Bee): 32 Chromosomes (Diploid / 2n).** Developed via standard zygotic combination of male and female gametes.
- **Y (Drone Bee Origin / Parthenogenesis): 16 Chromosomes (Haploid / n).** Developed from unfertilized eggs via parthenogenesis, retaining only a single set of maternal chromosomes.

Q6. Write any two differences between mitosis and meiosis cell division.

A side-by-side comparison reveals the following differences:

1. *Daughter Cell Output:* Mitosis yields **two genetically identical diploid (2n)** daughter cells, whereas meiosis yields **four genetically diverse haploid (n)** gametic cells.
2. *Biological Location:* Mitosis occurs in **somatic cells** for systemic growth and cellular repair, while meiosis occurs strictly within **germ cells** to form functional reproductive gametes.

Q7. Draw a probability table of the result obtained in sex determination.

Solution Table: Human sex determination tracks the chromosome configuration of the fertilizing spermatozoon:

- Union of X-sperm with X-egg yields **XX (Female Child)**.
- Union of Y-sperm with X-egg yields **XY (Male Child)**.

The genetic split results in an equal **50% Female (XX) / 50% Male (XY)** probability distribution across offspring.

Q8. Mention two functions of blood in the human body.

1. **Systemic Transport:** Delivers respiratory oxygen from the lungs to target tissues and returns carbon dioxide to be exhaled.
2. **Immune Shielding & Coagulation:** White blood cells neutralize foreign pathogens, while platelets coordinate coagulation at injury sites to prevent blood loss.

Q9. Identify the medicinal plant shown in the figure (Cinchona) and write its two uses.

- **Identification:** The plant is **Cinchona** (*Cinchona officinalis*).
- **Two Uses:** 1. The dried bark yields **quinine**, a primary pharmaceutical compound used to treat malaria. 2. It serves as an additive in digestive tonics to stimulate appetite and ease muscle cramps.

Q10. What is acceleration due to gravity? What is its value at the Earth's poles?

- **Definition:** Acceleration due to gravity (g) is the uniform acceleration experienced by a freely falling body solely due to the gravitational pull of a celestial body.
- **Polar Value:** At the Earth's poles, it reaches its maximum value of roughly **9.83 m/s^2** because the polar radius is shorter ($g \propto 1/R^2$).

Q11. If the given gravitational force between A and B is F , what is the gravitational force between B and C in terms of F ?

Solution: Following standard system metrics where C possesses three times the mass of A ($M_C = 3M_A$) over an identical distance d :

1. Force A-B: $F = (G \cdot M_A \cdot M_B) / d^2$.
2. Force B-C: $F' = (G \cdot M_B \cdot M_C) / d^2 = (G \cdot M_B \cdot 3M_A) / d^2$.
3. Substitution: $F' = 3 \cdot [(G \cdot M_A \cdot M_B) / d^2] = 3F$.

Q12. Write any two differences between an electric motor and a generator.

1. **Energy Conversion:** An electric motor transforms **electrical energy into mechanical energy**, whereas a generator transforms **mechanical energy into electrical energy**.
2. **Governing Law:** Motors leverage the **magnetic force on current-carrying conductors** (Lorentz Force), while generators operate via **electromagnetic induction** (Faraday's Law).

Q13. If the load transmission station at Dhalkevar transmits electricity at 400 KV, which type of transformer is used? Describe its role.

Answer: A **Step-up Transformer** is used. It steps up low generation voltages to an ultra-high 400 KV transmission level. This keeps transmission current low, minimizing power loss (I^2R) across long national grid networks.

Q14. A zinc plate is kept in a blue copper sulphate solution. The color fades, and small holes appear in the plate. State the reason and write the chemical equation.

Reason: Zinc is more reactive than copper and displaces copper ions from the solution. The blue Cu^{2+} ions reduce into brown copper metal deposit, causing the color to fade. The zinc plate dissolves as it oxidizes into Zn^{2+} ions, creating visible holes.

Equation: $\text{Zn (s)} + \text{CuSO}_4 \text{ (aq)} \rightarrow \text{ZnSO}_4 \text{ (aq)} + \text{Cu (s)}$ [Displacement Reaction].

Q15. A monohydric alcohol is used as a thermometric liquid and a disinfectant for wounds.

(a) Write its structural formula.

(b) Write the molecular formula of the fourth member of this homologous series.

Answer: The compound described is Ethanol ($\text{C}_2\text{H}_5\text{OH}$).

(a) Structural layout contains a two-carbon chain bound to a hydroxyl group: $\text{CH}_3\text{-CH}_2\text{-OH}$.

(b) The generic formula for alcohols is $\text{C}_n\text{H}_{2n+1}\text{OH}$. For the 4th member ($n=4$), the molecular formula of Butanol is **$\text{C}_4\text{H}_9\text{OH}$** .

Q16. How is bacterial growth inhibited by salting or adding sugar to food materials?

Answer: High salt or sugar concentrations create a highly **hypertonic environment** around bacterial cells. This triggers **exosmosis**, drawing water out through the bacteria's semi-permeable membranes. The resulting cellular dehydration (plasmolysis) halts metabolic activity and reproduction, preserving the food.

Section 'D' — Long Answer Type Questions (7 × 4 = 28 Marks)

Q17. Today, Facebook Messenger is widely used to communicate via social media.

(a) Name two other applications that function like Messenger.

(b) Write two points on how you can use these applications to improve your studies.

Answer:

(a) Alternative messaging apps include **WhatsApp Messenger** and **Viber**.

(b) *Educational Uses:*

1. **Collaborative Study Networks:** Students can form dedicated chat groups to share study guides, homework files, and science diagrams instantly.
2. **Real-Time Peer Support:** These apps enable instant voice and video calls to clarify complex concepts and troubleshoot difficult homework problems with classmates.

Q18. Artificial Insemination (AI) and In-Vitro Fertilization (IVF) are assisted reproductive technologies.

(a) 'AI technology helped to bring happiness to the farmer.' Justify with two opinions.

(b) 'IVF is proved to be a boon for childless couples.' Present two reasons.

Answer:

(a) *AI Benefits for Farmers:*

1. **Cost-Effective Breeding:** Farmers can upgrade their herds with premium pedigree genetics (such as high-yield Murrah strains) without the expense of keeping live bulls.
2. **Disease Management:** AI limits the spread of infectious reproductive diseases, ensuring healthier livestock and more predictable dairy production.

(b) *IVF Benefits for Childless Couples:*

1. **Bypasses Structural Obstacles:** IVF allows successful fertilization outside the body, bypassing blocked or damaged fallopian tubes.
2. **Addresses Severe Male Infertility:** Specialized variations like ICSI can achieve direct fertilization even with very low sperm counts, helping couples conceive successfully.

Q19. Write any four functions of plasma.

Answer:

1. **Nutrient Transportation:** Circulates glucose, amino acids, and lipids absorbed from the digestive system to cells across the body.
2. **Waste Elimination:** Transports cellular metabolic waste products, such as urea and carbon dioxide, to excretory organs for removal.
3. **Hormonal Distribution:** Carries essential regulatory hormones from endocrine glands directly to their target sites.
4. **Osmotic and Thermal Balance:** Uses plasma proteins (like albumin) to regulate blood pressure and body temperature.

Q20. Truck heat engines are cooled using water.

(a) Why is water used for this purpose?

(b) Numerical: Aakriti mixed 12 kg of water at 88°C (Bucket A) with 24 kg of water at 13°C (Bucket B) into Bucket C. Calculate the final equilibrium temperature.

Answer:

(a) Water is an ideal coolant because it has an exceptionally high **specific heat capacity (4200 J/kg°C)**. This allows it to absorb a substantial amount of engine heat with minimal temperature rise.

(b) *Numerical Calculation:*

By the Principle of Calorimetry: Heat Lost by Hot Water = Heat Gained by Cold Water

$$m_1 \cdot c \cdot (T_1 - \theta) = m_2 \cdot c \cdot (\theta - T_2)$$

Since c is identical, it cancels out on both sides:

$$12 \times (88 - \theta) = 24 \times (\theta - 13)$$

$$1 \times (88 - \theta) = 2 \times (\theta - 13) \text{ [Dividing both sides by 12]}$$

$$88 - \theta = 2\theta - 26$$

$$114 = 3\theta \rightarrow \theta = 114 / 3 = \mathbf{38^\circ C}.$$

Conclusion: The final equilibrium temperature of the mixture in bucket C is **38°C**.

Q21. Complete the schematic ray diagram for an object PQ placed between F and 2F of a convex lens and list its image characteristics.

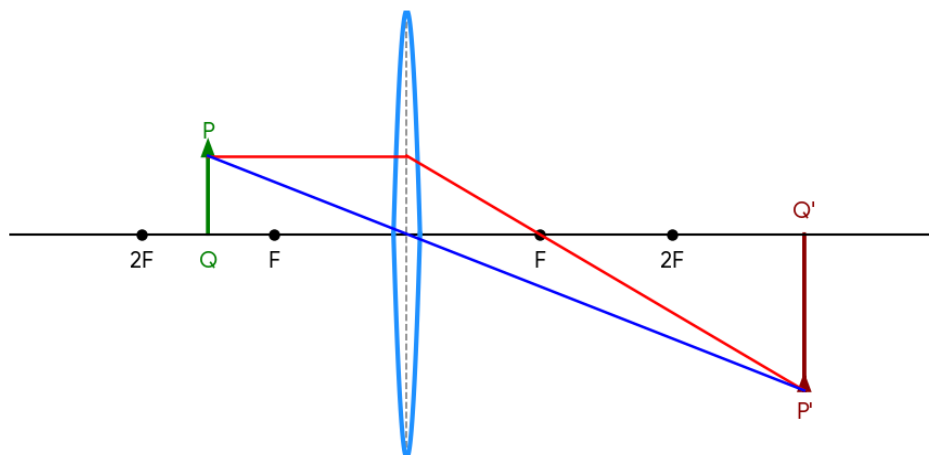


Image Characteristics:

1. The image is **Real and Inverted**.
2. It is **Magnified** (larger than the object) and forms beyond 2F on the opposite side.

Device Application: This configuration is used in **Film Projectors** to project an enlarged, real image of a small film slide onto a distant screen.

Q22. Answer the following based on periodic trends:

(a) Name the element with configuration: $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$.

(b) Name an alkali metal other than sodium.

(c) Name a metalloid other than silicon.

(d) Name an alkaline earth metal other than magnesium.

Answer:

(a) **Potassium (K)** [Atomic Number = 19]

(b) **Lithium (Li)** (or Rubidium, Caesium)

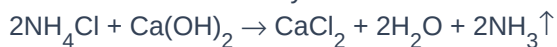
(c) **Germanium (Ge)** (or Boron, Arsenic)

(d) **Calcium (Ca)** (or Beryllium, Barium)

Q23. Detail the laboratory preparation of ammonia gas and explain how it is tested.

Answer:

• *Chemical Principle:* Ammonia (NH_3) is prepared in the laboratory by heating a 2:1 ratio mixture of Ammonium Chloride and Calcium Hydroxide:



• *Collection Method:* The gas is passed through a lime tower containing Calcium Oxide (CaO) to remove moisture, and then collected in an inverted gas jar via the **downward displacement of air** because it is lighter than air.

• *Testing Procedure:* Bring a piece of **moist red litmus paper** to the mouth of the jar. Because ammonia is alkaline, the paper will instantly turn **blue**. Alternatively, introducing a glass rod dipped in concentrated HCl will produce dense white fumes of Ammonium Chloride (NH_4Cl).